why 16 and not 4?

prof. dr. nicolas meseth

1. what the numbers say
2. what decides your alphabet
3. what it costs

part 1

what the numbers say

the lamp is as bright as ever, so why does its beam end so soon?

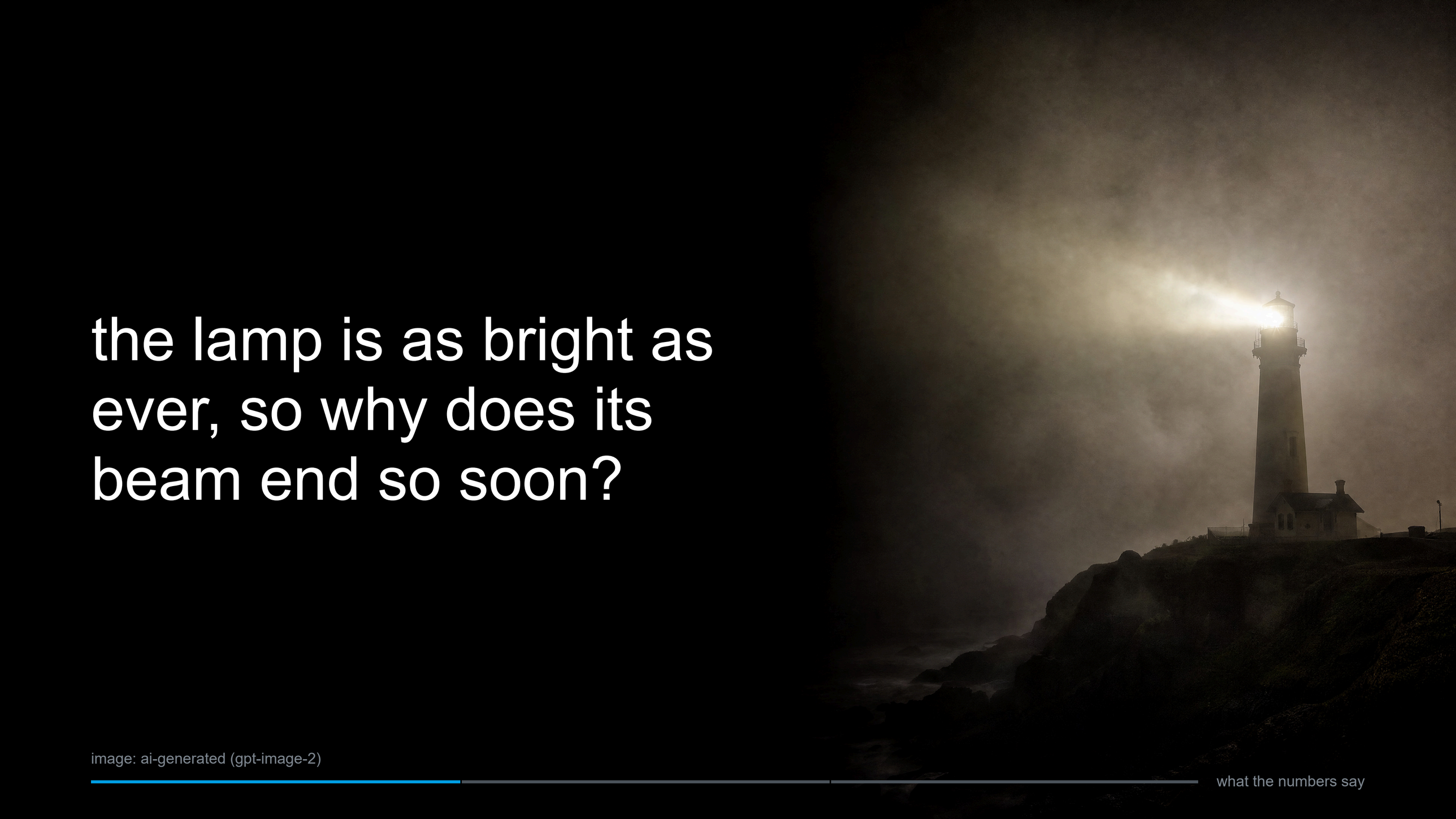


image: ai-generated (gpt-image-2)

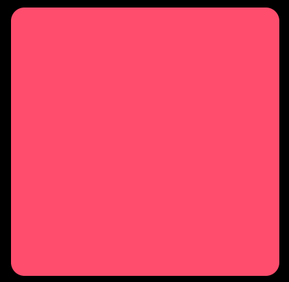
same hardware, different alphabets

team	alphabet	hit rate	what they did about noise
team a	16	98 %	cardboard tube, fixed distance, mean of 8
team b	8	96 %	longer integration time, shielded from the window
team c	8	91 %	picked colours further apart
team d	4	100 %	nothing, played it safe
team e	4	82 %	nothing, and it still failed

with the same hardware and task, the alphabets still differ by a factor of four.

example values; add your own team as a sixth row

ten readings of one colour

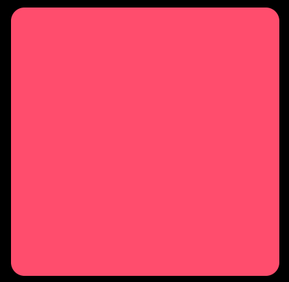


LED red
ten readings

what the sensor reports

214

ten readings of one colour



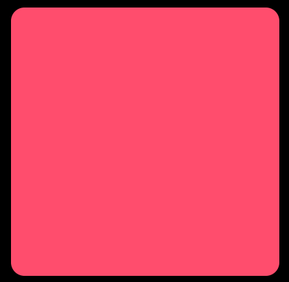
LED red
ten readings

what the sensor reports

214 211 218 209 216 213 220 207 215 212

nothing changed between the ten readings, and still every number is different.

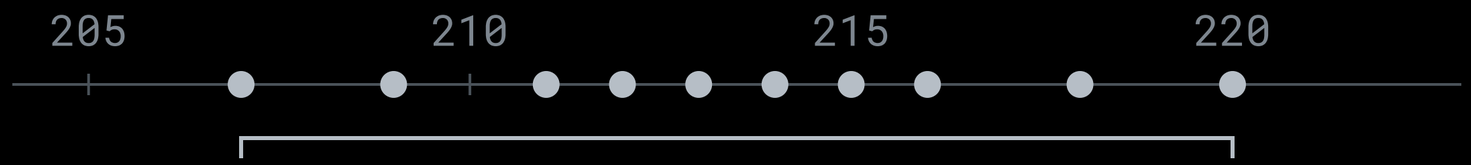
ten readings of one colour



LED red
ten readings

what the sensor reports

214 211 218 209 216 213 220 207 215 212



spread: 207 to 220

nothing changed between the ten readings, and still every number is different.

it is not the code that decides,
it is the scatter.

same sensor, same LED, same room: what differs is how far the readings move while nothing else moves.

part 2

what decides your alphabet

the same signal, averaged over longer windows

raw signal  288 readings

the same signal, averaged over longer windows

raw signal



288 readings

mean of 4



72 readings

the same signal, averaged over longer windows

raw signal



288 readings

mean of 4



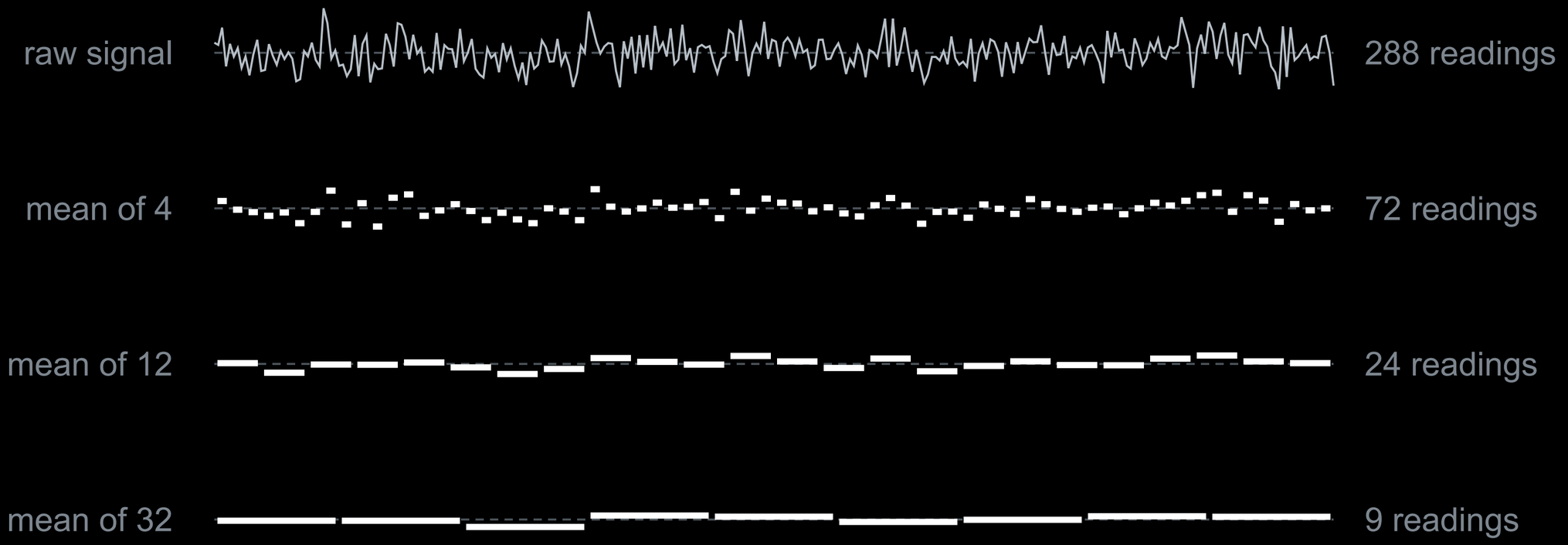
72 readings

mean of 12



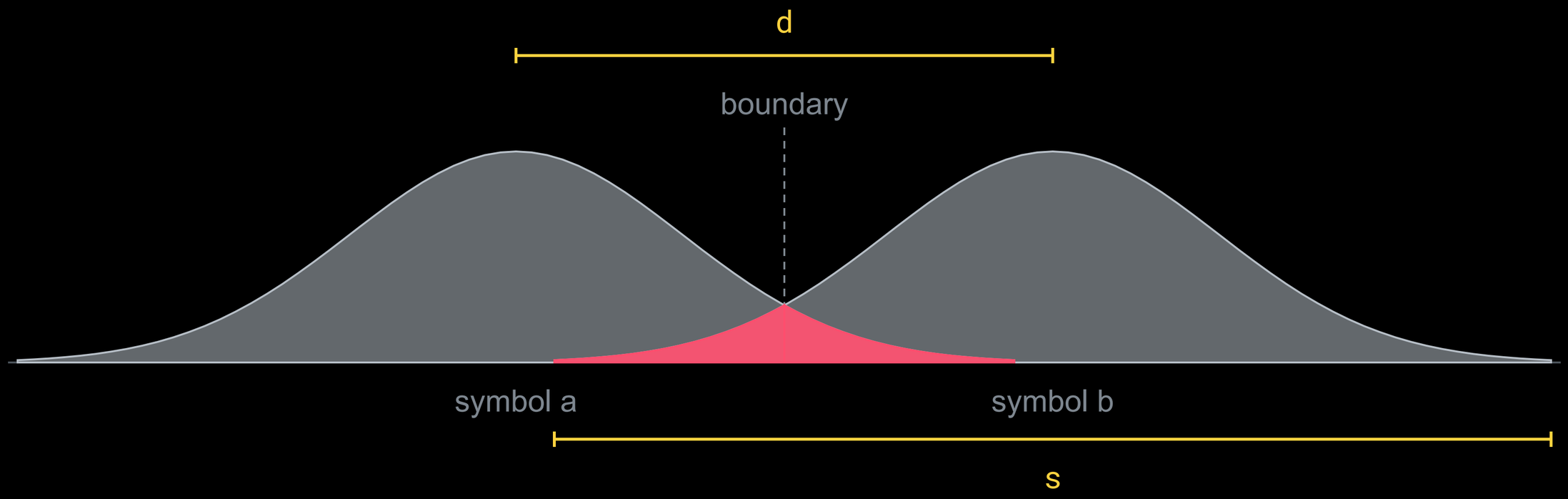
24 readings

the same signal, averaged over longer windows



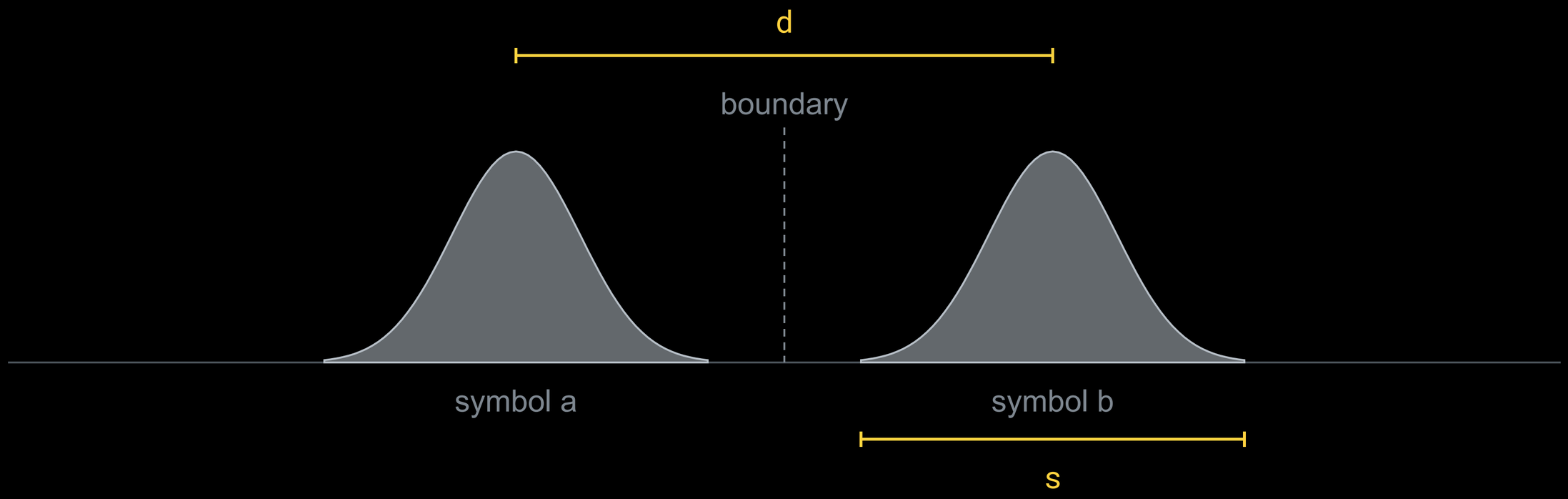
longer windows give calmer values, but fewer of them per second.

two symbols, two clouds



some readings land on the wrong side

two symbols, two clouds



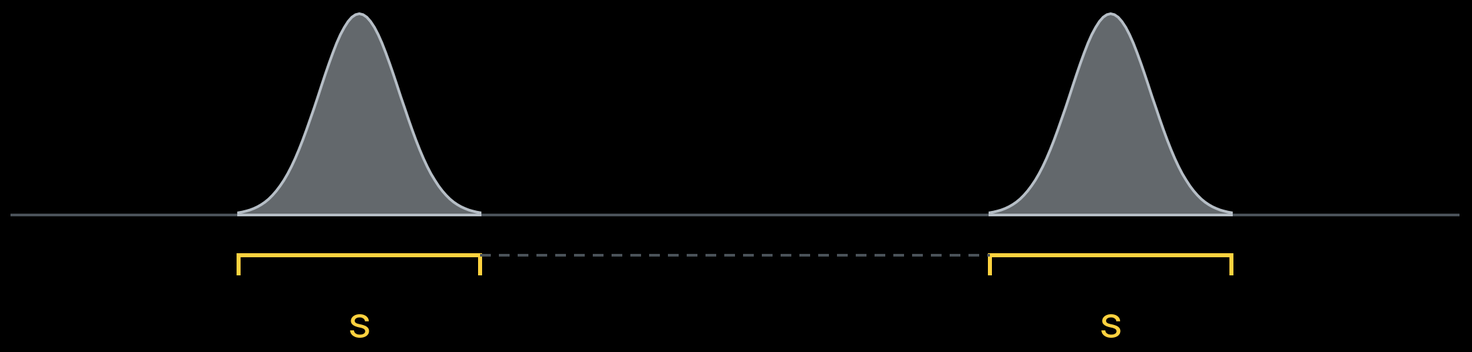
every reading lands on its own side

at the same distance, narrower clouds keep every reading on its own side.

d/s

d distance between two symbols

s spread of your readings



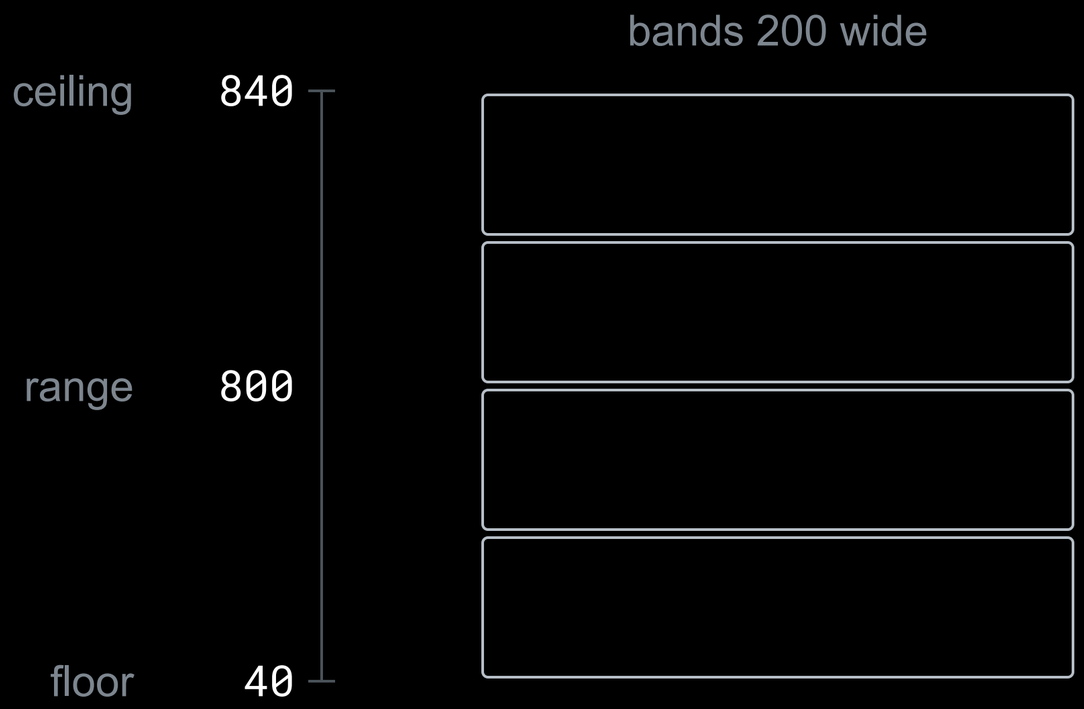
a gap between them, not a touch: that is what safely distinguishable means

your alphabet size measures nothing but this ratio.

how many symbols fit?



how many symbols fit?




```
800 / 200 = 4
```

how many symbols fit?

ceiling 840

range 800

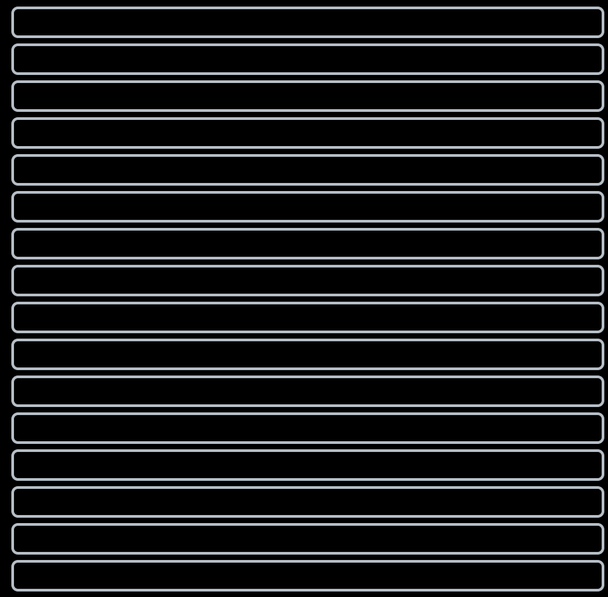
floor 40

bands 200 wide



800 / 200 = 4

bands 50 wide



800 / 50 = 16

the range divided by the width of one band gives the size of your alphabet.

eight symbols, two window lengths

short window



window four times as long



a dot outside the band of its symbol is read as another symbol

simulated readings; the trade-off is the one your own link shows

what decides your alphabet

eight symbols, two window lengths

short window



6 of 30 wrong

40 symbols/s

window four times as long



a dot outside the band of its symbol is read as another symbol

simulated readings; the trade-off is the one your own link shows

what decides your alphabet

eight symbols, two window lengths

short window



6 of 30 wrong

40 symbols/s

window four times as long



0 of 30 wrong

10 symbols/s

a dot outside the band of its symbol is read as another symbol

a longer window removes the errors, but it also cuts the symbols per second.

simulated readings; the trade-off is the one your own link shows

part 3

what it costs

bigger d

- **pick colours further apart**
  fewer symbols fit into the range
- **use fewer symbols**
  fewer bits per symbol
- **aim the light better**
  only craft

smaller s

- **longer integration time**
  fewer symbols per second
- **average several readings**
  fewer symbols per second
- **block ambient light**
  only craft
- **fix the geometry**
  only craft

everything else is paid for in bits per symbol or symbols per second

three of them cost nothing but craft, which is why self-built optics are allowed.

```
from lifi_hardware import sensor

def read_stable(n=10):
    values = [sensor.read()[3] for _ in range(n)]
    return sum(values) / n

print(read_stable())
```

the mean of ten is about three times calmer, but it takes ten times as long.



more symbols and faster symbols both spend the same margin, and you only have one.

one honest sentence

`we believed:` more gain gives us more colours

`we measured:` the scatter grew too, and we lost two symbols

an example from a log of mistakes: write your own pair for one suggestion of your assistant, the prediction before you measure and the result after it.

only a measurement on your own link can settle a claim about it.

the fog stays, so
measure how far you
can see.

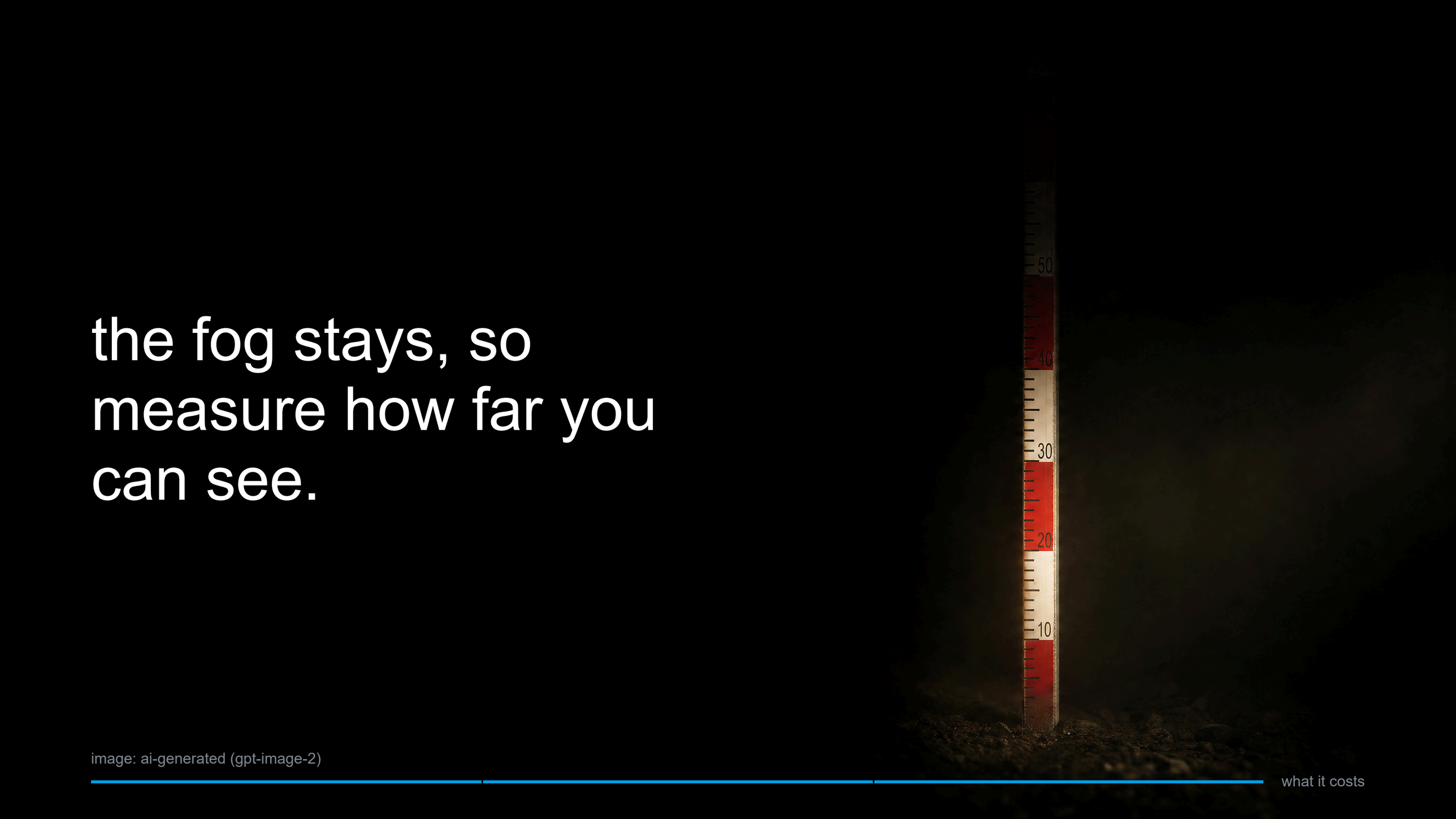


image: ai-generated (gpt-image-2)